



Course: B Tech

Semester: 3

Prerequisite: Basic knowledge of programming

Rationale: The practical component reinforces Core Java concepts through hands-on programming. It helps students apply OOP, collections, exception handling, file I/O, multithreading, and Java 8 features to solve real-world problems, preparing them for advanced development and industry roles.

Teaching and Examination Scheme										
Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	0	0	3	20	20	-	60	-	100

SEE - Semester End Examination, CIA - Continuous Internal Assessment (It consists of Assignments/Seminars/Presentations/MCQ Tests, etc.)

Course Content		W - Weightage (%), T - Teaching hours	
Sr.	Topics	W	T
1	Overview of OOP and Java: Programming Paradigms, Procedural vs Object-Oriented Programming, OOP Features, History of Java, Feature of Java, JVM Architecture, JRE, JDK, Java Program Structure, Compilation and Execution Process, Java Environment Setup, JDK Installation, PATH and CLASSPATH, Introduction to IDEs.	10	4
2	Java Language Basics: Variables, Scope of Variables, Primitive Data Types, Reference Data Types, Type Casting, Wrapper Classes, Autoboxing and Unboxing, Arithmetic Operators, Relational Operators, Logical Operators, Assignment Operators, Unary Operators, Bitwise Operators, Operator Precedence, if Statement, if-else Statement, Nested if, switch Statement, for Loop, while Loop, do-while Loop, for - Each Loop, break Statement, continue Statement.	15	7
3	Arrays and String Arrays, One-Dimensional Arrays, Multi-Dimensional Arrays, Array Operations, String Class, StringBuffer, StringBuilder, String Comparison, String Manipulation Methods	17	6
4	Object-Oriented Programming (OOP) Classes and Objects, Constructors, Access Specifiers, Encapsulation, Inheritance, Types of Inheritance, super Keyword, Polymorphism, Method Overloading(Compile time Polymorphism), Method Overriding (Runtime Polymorphism), Abstraction, Abstract Classes, Interfaces, Functional Interfaces, this Keyword, final Keyword, Object Class and its Methods.	30	12
5	Exception Handling Errors and Exceptions, Checked Exceptions, Unchecked Exceptions, try-catch Block, Multiple catch Blocks, finally Block, throw Keyword, throws Keyword, Custom Exceptions, Exception Handling, Best Practices	15	6
6	Collection Framework in Java Introduction to Collections Framework, List Interface, ArrayList, LinkedList, Set Interface, HashSet, TreeSet, Map Interface, HashMap, TreeMap, Iterators, Generics.	10	5
7	Advanced Foundations and Java Frameworks Process Vs Thread, Thread Creation, Thread Life Cycle, Multithreading Concepts, Synchronization, Inter-thread Communication, File Handling, File Class, Byte Streams, Character Streams, Buffered Streams, Serialization, Deserialization, Lambda Expressions, Stream API, Optional Class.	8	5
Total		100	45



Reference Books	
1.	Herbert Schildt, Java: The Complete Reference, McGraw-Hill.
2.	Cay S. Horstmann, Core Java – Volume I & II, Pearson.
3.	Joshua Bloch, Effective Java, Addison-Wesley.
4.	Kathy Sierra & Bert Bates, Head First Java, O'Reilly.

Course Outcome	
After Learning the Course the students shall be able to:	
1. Explain Java architecture, JVM working, and program execution process. 2. Apply core programming constructs including data types, control statements, and operators. 3. Implement object-oriented programming concepts to design modular applications. 4. Utilize collections, exception handling, file I/O, and multithreading in application development	